



2022 Mathematics Standard Preliminary HSC Examination Marking Guidelines

Section I

Multiple Choice Answer Key

Question	Answer
1	C
2	B
3	A
4	B
5	B
6	C
7	C
8	A
9	A
10	C

Question 1

$$s = 3.11 \times 4 + \frac{1}{2} \times 4.52 \times 4^2$$

$$\therefore s = 48.60$$

Answer: C

Question 2

$$\text{Accuracy} = 0.1$$

$$\text{Absolute error} = 0.05$$

$$\begin{aligned} \text{Percentage error} &= \frac{0.05}{12.4} \times 100 \\ &= 0.403\% \end{aligned}$$

Answer: B

Question 3

Option A is the only graph showing a straight line with y-intercept 1 and a positive gradient $\left(\frac{3}{4}\right)$.

Answer: A

Question 4

Only the sides of the triangle in Option B satisfy Pythagoras' Theorem.

$$\text{i.e. } \sqrt{8^2 + 15^2} = 17$$

Answer: B

Question 5

Let the cost price of the vase be X .

$$1.10 \times X = 16.50$$

$$X = \frac{16.50}{1.10}$$

$$\therefore X = 15$$

Answer: B

Question 6

This is an example of systematic sampling.

Answer: C

Question 7

If Robbie's score is removed then the mean will increase and the range will decrease.

Answer: C

Question 8

$$\frac{6}{18} \times 3000 = 1000 \text{ eggs.}$$

Answer: A

Question 9

$$100 \text{ m}^2 \xrightarrow{\div 1000^2} 0.0001 \text{ km}^2$$

Answer: A

Question 10

$$500 = 500 \times 7\% \times N$$

$$N = \frac{500}{500 \times 7\%}$$

$$N = 14.28$$

Therefore, it takes 15 years for the initial \$500 invested to double its value.

Answer: C

Section II**Question 11**

Criteria	Marks
• Provides the correct solution	2
• Correctly substitutes given values	1

Let w = weight in kg

$$110 = \frac{w \times 500}{70}$$

$$w = \frac{110 \times 70}{500}$$

$$\therefore w = 15.4 \text{ kg}$$

Question 12

Criteria	Marks
• Provides the correct solution	2
• Makes progress towards solution	1

$$5x + 20 = x - 2$$

$$4x = -22$$

$$\therefore x = -\frac{11}{2} \text{ OR } -5.5$$

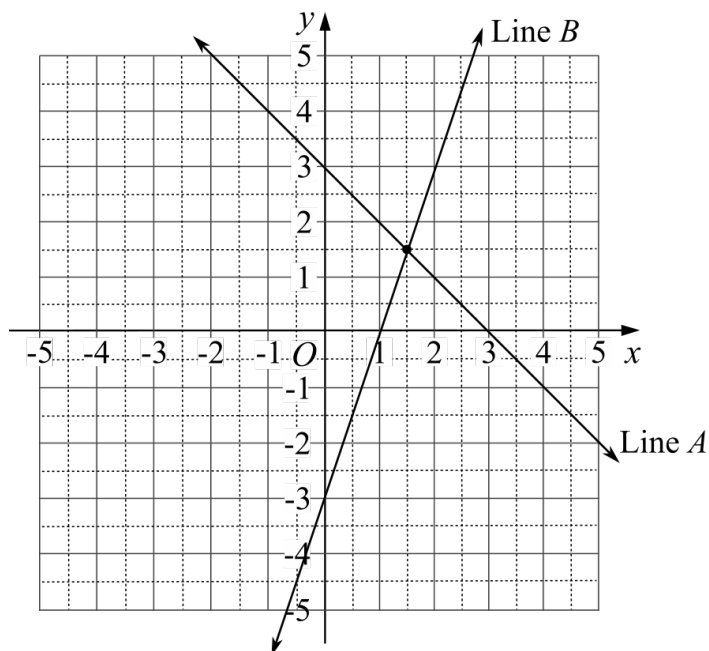
Question 13 (a)

Criteria	Marks
Provides the correct answer	1

$$m = -1$$

Question 13 (b)

Criteria	Marks
Provides the correct answer	2
Correct gradient OR y-intercept	1

**Question 13 (c)**

Criteria	Marks
Provides the correct answer	1

$(1.5, 1.5)$

Question 14 (a)

Criteria	Marks
• Provides the correct solution	2
• Correctly uses the formula for the circumference of a circle	1

$$\begin{aligned}P &= 2 \times 2\pi r \\&= 2 \times 2\pi \times 4 \\&= 16\pi \\\therefore P &= 50.3 \text{ m}\end{aligned}$$

Question 14 (b)

Criteria	Marks
• Provides the correct solution	2
• Correctly uses the formula for the area of a circle, or equivalent merit	1

$$\begin{aligned}A &= A_{\text{square}} + 2 \times A_{\text{circle}} \\&= 8^2 + 2 \times \pi \times 4^2 \\\therefore A &= 164.5 \text{ m}^2\end{aligned}$$

Question 15 (a)

Criteria	Marks
Provides the correct answer	1

$$35 \times (14.95 - 3.99) = \$383.60$$

Question 15 (b)

Criteria	Marks
Provides the correct solution	2
Finds the total amount of money saved on electricity each day, or equivalent merit	1

$$8 \text{ watts} \times 35 \text{ lights} \times 4 \text{ hours} = 1120 \text{ watts}$$
$$\Rightarrow 1.12 \text{ kW saved each day.}$$

$$1.12 \times 0.27 = 0.3024 \text{ dollars saved each day.}$$

$$383.60 \div 0.3024 = 1268.52.$$

$\therefore$ it takes 1269 days.

Question 16 (a)

Criteria	Marks
Provides the correct answer	1

$$72^{\circ} + 13^{\circ} = 85^{\circ}$$

Question 16 (b)

Criteria	Marks
Provides the correct solution	3
Correctly states the time OR day when Kate receives the email, or equivalent merit	2
Finds the time difference, or equivalent merit	1

$$\frac{85^{\circ}}{15^{\circ}} = 5 \text{ h } 40 \text{ min}$$

Kate is 5 hours and 40 minutes behind Liam.

Therefore, she receives the email at 2:20 pm on Sunday.

Question 17

Criteria	Marks
Provides the correct solution	3
Correctly solves for the value of n, or equivalent merit	2
Correctly substitutes given values into $S = V_0 - Dn$	1

$$S = V_0 - Dn$$

$$20\,000 = 25\,870 - 700n$$

$$n = 8.386$$

Therefore, the jet ski's salvage value will be \$20 000 in the 9th year.

Question 18

Criteria	Marks
• Provides the correct solution	5
• Finds the amount of double time required	4
• Deduces the amount to be earned by working at double time, or equivalent merit	3
• Finds the total amount that Jessica needs to earn each week, or equivalent merit	2
• Finds how much needs to be saved each week, or equivalent merit	1

Jessica needs to save $18\,200 \div 52 = \$350$ per week.

Hence, she needs to earn $\$875 + \$350 = \$1225$ per week.

$\$1225 - (31 \times \$35) \Rightarrow$ She needs to earn \$140 from double time.

i.e. $140 = n \times (31 \times 2)$

$$n = \frac{140}{31 \times 2}$$

$$\therefore n = 2.26 \text{ hours} = 2 \text{ hours and } 16 \text{ minutes}$$

Therefore, Jessica needs to work 37 hours and 16 minutes each week.

Question 19 (a)

Criteria	Marks
Provides the correct answer	1

$$113\,000 - 6700 - 3000 \\ = \$103\,300$$

Question 19 (b)

Criteria	Marks
Provides the correct solution	2
Correctly calculates medicare levy, or equivalent merit	1

$$\text{Medicare levy} = 2\% \times 103\,300 = \$2066$$

$$\text{Tax Payable} = 5092 + 0.325(103\,300 - 45\,000) = \$24\,039.50$$

$$\text{Total tax payable} = 2066 + 24\,039.50 = \$26\,105.50$$

Question 19 (c)

Criteria	Marks
Provides the correct solution	2
Correctly calculates difference between Total Tax Payable and annual PAYG	1

$$\text{PAYG} = 26 \times 1216 = \$31\,616$$

Therefore Stephen should be refunded the amount of
 $31\,616 - 26\,105.50 = \$5510.50$

Question 20

Criteria	Marks
Provides the correct solution	2
Correctly calculates the total number of students, or equivalent merit	1

$$\frac{150}{800} \times 200 = 37.5$$

Therefore 38 responses should be allocated to Year 10 students.

Question 21 (a)

Criteria	Marks
Provides the correct answer	1

$$\frac{15 + 19}{2} = 17$$

Question 21 (b)

Criteria	Marks
Provides the correct answer	2
Correctly identifies the lower and upper quantiles, or equivalent merit	1

Lower quartile: $Q_1 = 10$

Upper quartile: $Q_3 = 31$

Interquartile range (IQR) = $31 - 10 = 21$

Question 21 (c)

Criteria	Marks
Provides the correct answer	2
Correctly identifies the lower and upper bounds, or equivalent merit	1

$$\begin{aligned}\text{Lower bound} &= Q_1 - 1.5 \times \text{IQR} \\ &= 10 - 1.5 \times 21 \\ &= -21.5\end{aligned}$$

$$\begin{aligned}\text{Upper bound} &= Q_3 + 1.5 \times \text{IQR} \\ &= 31 + 1.5 \times 21 \\ &= 62.5\end{aligned}$$

No scores are less than -21.5 or greater than 62.5 . Therefore there are no outliers.

Question 22 (a)

Criteria	Marks
• Provides the correct answer	1

$$86 + 62 + 28 + 16 + 8 = 200$$

Question 22 (b)

Criteria	Marks
• Provides the correct answer	2
• Finds the total of pocket and collar defects, or equivalent merit	1

$$88\% - 43\% = 45\%$$

OR

$$\frac{62 + 28}{200} \times 100\% = 45\%$$

Question 23 (a)

Criteria	Marks
• Provides the correct solution	2
• Some answers are correct	1

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

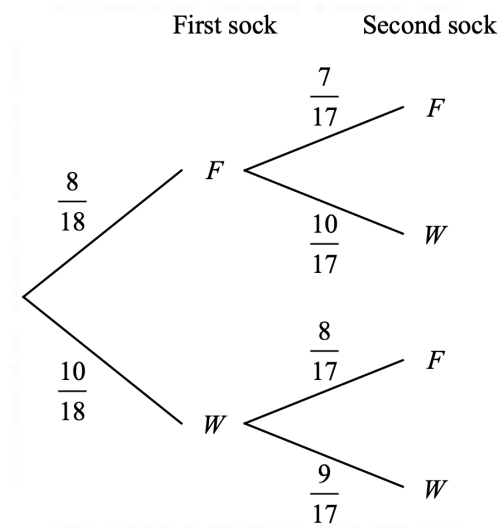
Question 23 (b)

Criteria	Marks
• Provides the correct solution	2
• Correctly finds the number of outcomes that are great than 6, or equivalent merit	1

$$P(6) = \frac{21}{36} = \frac{7}{12}$$

Question 24 (a)

Criteria	Marks
• Provides the correct solution	2
• Correctly identifies some probabilities	1

**Question 24 (b)**

Criteria	Marks
• Provides the correct solution	2
• Correctly calculates the some of a multi-stage event, or equivalent merit	1

$$\begin{aligned}
 P(FF) + P(WW) &= \left(\frac{8}{18} \times \frac{7}{17} \right) + \left(\frac{10}{18} \times \frac{9}{17} \right) \\
 &= \frac{73}{153}
 \end{aligned}$$

Question 25 (a)

Criteria	Marks
• Provides the correct solution	3
• Finds the curved surface area of the cylindrical part, or equivalent merit	2
• Finds the area of the hemisphere, or equivalent merit	1

$$\begin{aligned}
 SA &= \frac{1}{2} \times 4\pi r^2 + 2\pi rh + \pi r^2 \\
 &= 3\pi r^2 + 2\pi rh \\
 &= 3\pi \times 3^2 + 2\pi \times 3 \times 10 \\
 \therefore SA &= 237.32 \text{ m}^2
 \end{aligned}$$

Question 25 (b)

Criteria	Marks
• Provides the correct solution	3
• Finds the total volume of the water tank	2
• Finds the volume of the hemisphere OR cylinder, or equivalent merit	1

$$\begin{aligned}
 V &= \frac{1}{2} \times \frac{4}{3} \pi r^3 + \pi r^2 h \\
 &= \frac{2}{3} \pi \times 3^3 + \pi \times 3^2 \times 10 \\
 \therefore V &= 339.3 \text{ m}^3
 \end{aligned}$$

Therefore the water tank holds 339 kL (to the nearest kL).

Question 25 (c)

Criteria	Marks
• Provides the correct solution	3
• Correct approximation for the area of the farm, or equivalent merit	2
• One correct application of the Trapezoidal Rule, or equivalent merit	1

$$A_{\text{total}} = A_1 + A_2$$

$$A_1 = \frac{175}{2}(200 + 180)$$

$$= 33\,250 \text{ m}^2$$

$$A_2 = \frac{175}{2}(180 + 0)$$

$$= 15\,750 \text{ m}^2$$

$$A_{\text{total}} = 33\,250 + 15\,750$$

$$= 49\,000 \text{ m}^2$$

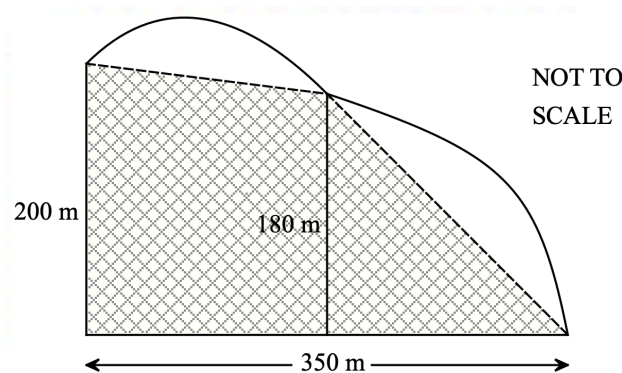
$$\text{Water required} = 339\,000 \div 49\,000 = 6.92 \text{ L/m}^2$$

Question 25 (d)

Criteria	Marks
• Provides the correct solution, with justification	1

The area approximated by the Trapezoidal Rule is an underestimate of the actual area of the farm.

By drawing the trapezia (shaded) used in the approximation, it is evident that there is some farmland unaccounted for.



Mathematics Standard Preliminary HSC Examination Mapping Grid			
Question	Parts	Marks	Content
1	MC	1	MS-A1 Formulae and Equations
2	MC	1	MS-M1 Applications of Measurement
3	MC	1	MS-A2 Linear Relationships
4	MC	1	MS-M1 Applications of Measurement
5	MC	1	MS-F1 Money Matters
6	MC	1	MS-S1 Data Analysis
7	MC	1	MS-S1 Data Analysis
8	MC	1	MS-S2 Relative Frequency and Probability
9	MC	1	MS-M1 Applications of Measurement
10	MC	1	MS-F1 Money Matters
11	1	2	MS-A1 Formulae and Equations
12	1	2	MS-A1 Formulae and Equations
13	3	4	MS-A2 Linear Relationships
14	2	4	MS-M1 Applications of Measurement
15	2	3	MS-M1 Applications of Measurement
16	2	4	MS-M2 Working with Time
17	1	3	MS-F1 Money Matters
18	1	5	MS-F1 Money Matters
19	3	5	MS-F1 Money Matters
20	1	2	MS-S1 Data Analysis
21	3	5	MS-S1 Data Analysis
22	2	3	MS-S1 Data Analysis
23	2	4	MS-S2 Relative Frequency and Probability
24	2	4	MS-S2 Relative Frequency and Probability
25	4	10	MS-M1 Applications of Measurement